

Bachelor in Computer Applications

Course Title: **Discrete Structure**

Code No: Semester: **2**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Explain direct proof, indirect proof, and proof by contradiction. Use direct proof to show that "If n is an odd integer, then n^3 is an odd integer". Also use indirect proof to show that "If n is an integer and n^3 then n is odd".
2. What is linear nonhomogeneous recurrence relation of degree k with constant coefficients? Find all the solutions of the recurrence relation $a_{n+k} = a_n + n$. Also find the solution of the relation with initial condition $a_1 = 1$.
3. Define spanning tree and minimum spanning tree with suitable example. Use Kruskal's algorithms to find minimum spanning tree in the given graph.
4. What is tautology? Show $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.
5. Define cartesian product. Find A^3 for the set $A = \{a, b, c\}$.
6. How can you represent relations using matrices? Suppose that $A = \{1, 2, 3\}$ and $B = \{1, 2\}$. Let R be the relation from A to B containing (a, b) if $a \in A$, $b \in B$, and $a > b$. What matrix representing R if $a_1 = 1$, $a_2 = 2$, $a_3 = 3$, and $b_1 = 1$ and $b_2 = 2$?
7. Use mathematical induction to show that the sum of first n positive integers is $n(n+1)/2$.
8. What is congruent modulo? Determine whether 20 is congruent to 8 modulo 6 and 25 is congruent to 17 modulo 5.
9. Explain trial division with example? Using trial division, show that 101 is prime.
10. Explain product rule. How many strings are there of four lowercase letters that have the letter x in them?
11. What is graph? Explain simple graph and pseudograph with example.
12. What is Euler path? Compare it with Hamilton path.